CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 8: Predicting What Comes Next? Exploring Sequences and Progressions

Every formula and result of the chapter on one sheet

1	Explicit rule	t_n written in terms of n
2	Recursive rule	t_n written in terms of earlier terms, with the starting term(s) given
3	Virahānka–Fibonacci sequence	$V_1 = 1, V_2 = 2, V_n = V_{n-1} + V_{n-2}$
4	General AP	$a,\ a + d,\ a + 2d,\ \ldots$
5	nth term of an AP	$t_n = a + (n - 1)d$
6	Recursive rule of an AP	$t_1 = a, t_n = t_{n-1} + d$
7	Common difference	$d = t_2 - t_1$
8	Sum of the first n natural numbers	$\frac{n(n+1)}{2}$
9	nth triangular number	$\frac{n(n+1)}{2}$
10	nth square number	$n^2 = 1 + 3 + 5 + \ldots + (2n - 1)$
11	General GP	$a,\ ar,\ ar^2,\ \ldots$
12	nth term of a GP	$t_n = ar^{n-1}$
13	Recursive rule of a GP	$t_1 = a, t_n = r \times t_{n-1}$
14	Common ratio	$r = \frac{t_2}{t_1}$
15	Sierpiński triangle, stage n	3^n triangles with total area $\left(\frac{3}{4}\right)^n$